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Complete transversals and the classification of singularities

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Abstract. The classification of map-germs (up to a variety of equivalences) has many applications in differential geometry, to the study of wavefronts and caustics and to bifurcation theory. In a previous paper the first and last authors together with C T C Wall, gave some useful criteria for map-germs to be finitely determined with respect to a wide range of equivalence relations. The results presented in this paper, used in conjunction with these determinacy techniques, provide a very efficient classification procedure. Moreover, we show that the algebraic criteria involved in these calculations may be reduced to finite-dimensional symbolic problems which may be performed by a computer.

AMS classification scheme number: 58C

In a previous paper Bruce *et al* [9] gave some useful criteria for map-germs to be finitely determined with respect to a wide range of equivalence relations. Indeed, the conditions given were necessary as well as sufficient, and so lead to a complete solution of the determinacy question. In this work a key part is played by the notion of a unipotent group. Indeed it seems reasonable that it will be easier to deal with groups of diffeomorphism-germs, whose linear parts form a unipotent group, than with the general case, for we are then essentially concerned with upper uni-triangular matrices, which form an affine space. It is a little more surprising that such groups are precisely what is needed to obtain an exact value for degrees of determinacy.

The importance of determinacy stems at least partly from its major role in the classification of map-germs. One way to proceed with the classification is inductively, classifying in turn all $(k+1)$-jets with a given k-jet, obtaining a classification tree, with various levels corresponding to the degree of the jets involved. A branch will terminate when we reach a k-jet which is k determined (with respect to whichever equivalence relation we are working with). Having solved the determinacy problem it is natural to turn to the other part of this classification procedure. How does one obtain, in an efficient manner, a list of all representatives of $(k+1)$-jets with respect to a given k-jet? In this paper we show that a useful procedure is available if we consider groups of jet-transformations whose

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linear parts are the identity. Such groups are unipotent and it is natural to seek an extension of this method to the case of general unipotent groups. If one refines the filtration by degree then one can obtain more general results which are also very useful.

There have been a number of other classification schemes which are, essentially, special cases of the one given here. In particular the spectral sequence methods of Arnold given in [2], which deals with the classification of certain types of functions up to right ($\mathcal{R}$) equivalence can be obtained using our methods, whilst the results of Dimca and Gibson [12, 13] on contact ($\mathcal{K}$) equivalence are of exactly the same type as those given here, so that our work is a direct generalization of theirs.

The results presented in this paper, used in conjunction with the determinacy techniques discussed in [9], provide a very efficient classification procedure. Moreover, the algebraic criteria involved in these calculations may be reduced to finite dimensional symbolic problems which may be performed by a computer. In fact, the calculations needed for a full classification are so extensive that this is the natural way to proceed. Computational methods suitable for performing the symbolic calculations central to classification and unfolding theory were developed in [17]; a computer package called `Transversal` was written by the second author using the symbolic algebra system `Maple`. This implementation of our results on a computer provides a very powerful tool, and we will devote some time to discussing the algorithm and giving some illustrative examples. Of course any algorithm can only produce a finite list of germs. However, it is often the initial part of the classifications which prove to be the most difficult. The computer calculations then produce stems, that is jets which are the initial parts of natural infinite families (see [23]). Once the stem is produced the 'theoretical' calculations needed to determine the infinite family are often straightforward. Most of the results below are valid in both the real smooth and complex analytic cases. We shall use the symbol $\boldsymbol{K}$ to denote both the real and complex numbers.

A number of substantial classifications have been carried out as a direct consequence of this work; see [6, 8, 15, 16, 25]. Ratcliffe has also used some of these methods in a number of her papers [27, 28].

Generalizations of the theory of complete transversals and applications to the recognition of singularities will appear in a separate article [7], currently in preparation.

1. Introductory material

Our main references for singularity theory will be [31] and [9]. We shall start our investigations with a few elementary lemmas concerning Lie group actions. The first is usually referred to in the literature as *Mather's lemma*.

Lemma 1.1. *Let G be a Lie group acting smoothly on a finite dimensional manifold V. Let X be a connected submanifold of V. Then X is contained in a single orbit of G if and only if*

(i) for each $x \in X$, the tangent space $T_x X \subset T_x(G \cdot x)$, and
(ii) $\dim T_x(G \cdot x)$ *is constant for all $x \in X$.*

For the proof see [21, lemma 3.1].

Lemma 1.2. *Let G be a connected Lie group acting smoothly on an affine space A; let W be a vector subspace of V_A. Then the action of G induces an action on A/W if and only if for all $x \in A$, $w \in W$ and $l \in LG$ we have*

$$l \cdot (x + w) - l \cdot x \in W.$$

For the proof see [9, lemma 2.2].

The following is the crucial result.

Proposition 1.3. *Let G be a Lie group acting smoothly on an affine space A, and let W be a subspace of V_A with*

$$LG \cdot (x + w) = LG \cdot x \tag{1}$$

for all $x \in A$ and $w \in W$. Then

(i) For any $x \in A$ we have

$$x + \{LG \cdot x \cap W\} \subset G \cdot x \cap \{x + W\}.$$

(ii) If $x_0 \in A$ and T is a vector subspace of W satisfying

$$W \subset T + LG \cdot x_0 \tag{2}$$

then for any $w \in W$ there exists $g \in G$, $t \in T$ such that $g \cdot (x_0 + w) = x_0 + t$.

Proof. The set $x + LG \cdot x \cap W$ is an affine subspace of A, with tangent space $LG \cdot x \cap W$. If $x + w \in x + LG \cdot x \cap W$ then

$$T_{x+w}(G \cdot (x + w)) = LG \cdot (x + w) = LG \cdot x$$

by condition (1) above. So the tangent spaces to the orbits through points of $x + LG \cdot x \cap W$ have constant dimension, and contain the tangent space to $x + LG \cdot x \cap W$. By Mather's lemma we deduce that

$$x + LG \cdot x \cap W \subset G \cdot x$$

and (i) follows.

To prove (ii) we note that

$$\begin{aligned}
\bigcup_{t \in T} G \cdot (x_0 + t) &\supset \bigcup_{t \in T} \{x_0 + t + LG \cdot (x_0 + t) \cap W\} \qquad \text{by (i)} \\
&= \bigcup_{t \in T} \{x_0 + t + LG \cdot x_0 \cap W\} \qquad \text{by (1)} \\
&= x_0 + T + LG \cdot x_0 \cap W \\
&= x_0 + (T + LG \cdot x_0) \cap W \qquad \text{since } T \subset W \\
&= x_0 + W \qquad \text{by (2).}
\end{aligned}$$

□

Proposition 1.3 part (ii) says that the transversal T to the orbit of x_0 meets each orbit through the affine subspace $x_0 + W$ of A. Consequently T is referred to as a *complete transversal*. This term will also be used to describe the affine subspace $x_0 + T$ or a basis for T.

In practice, condition (1) above is usually replaced by the sharper condition

$$l \cdot (x + w) = l \cdot x \tag{3}$$

for all $x \in A$, $w \in W$ and $l \in LG$. A correspondingly sharper result can then be deduced, namely:

Proposition 1.4. *Given the additional hypothesis (3) there exists a closed connected Lie subgroup H of G such that for any $x \in A$ we have*

$$x + \{LG \cdot x \cap W\} = H \cdot x.$$

(The subgroup H depends on $x + W$ but not on x itself.)

Proof. To prove this sharper assertion we set $L = \{l \in LG : l \cdot x \in W\}$; clearly $L \cdot x = LG \cdot x \cap W$. Moreover L and $L \cdot x$ are unaffected by the translation of x by an element of W. The group H is now constructed from L as in [9, lemma 4.3]. □

The previous two results apply to any Lie group acting smoothly on an affine space. However, all the applications that we have in mind are of algebraic groups over $\boldsymbol{R}$ or $\boldsymbol{C}$ acting algebraically on an affine space and in this situation a stronger result than proposition 1.4 can be established.

Proposition 1.5. *As usual let $\boldsymbol{K}$ be $\boldsymbol{R}$ or $\boldsymbol{C}$ and let G be an algebraic group over $\boldsymbol{K}$ acting on an affine space A. Let W be a subspace of V_A with*

$$l \cdot (x + w) = l \cdot x$$

for all $x \in A$, $w \in W$ and $l \in LG$. Then there exists a unipotent subgroup U of G such that

(i) $x + \{LG \cdot x \cap W\} = U \cdot x$ for any $x \in A$.

(ii) If $x_0 \in A$ and T is a vector subspace of W such that

$$W \subset T + LG \cdot x_0$$

then for any $w \in W$, there exist $u \in U$, $t \in T$ such that $u \cdot (x_0 + w) = x_0 + t$.

Proof. By proposition 1.4 there is a subgroup H of G such that $x + \{LG \cdot x \cap W\} = H \cdot x$ for any $x \in A$. We may suppose that H is algebraic, by the argument of [9, lemma 4.3] applied to the polynomial ϕ. But then, by [9, theorem 3.6] the unipotent radical U of H is transitive on $x + \{LG \cdot x \cap W\}$ for all $x \in A$, proving (i). Part (ii) now follows from proposition 1.3. □

It will be seen that in applications of this result one may as well restrict attention to unipotent groups, and this is what we do in the rest of the paper. We now illustrate the efficacy of these results in classification problems.

2. Classification problems

In this section we show how the complete transversal idea can be used to obtain efficient methods for classification. We begin with the most straightforward (and central) equivalence relations in singularity theory, and the filtration of germs by degree. We start by considering the group of equivalences with 1-jet the identity. This can be extended to larger subgroups (any unipotent jet group) provided that we use a finer filtration. Fewer *ad hoc* changes of co-ordinates are then required, but many more levels of classification are needed. Finally we deal with the weighted homogeneous filtrations whose importance has been highlighted by the work of Arnold, see [3].

2.1. The basic classification result

Definition 2.1. Let $\mathcal{G}$ be one of the groups $\mathcal{R}$, $\mathcal{L}$, $\mathcal{A}$, $\mathcal{C}$ or $\mathcal{K}$. The standard k-jet space $\mathcal{M}_n\mathcal{E}(n, p)/\mathcal{M}_n^{k+1}\mathcal{E}(n, p)$ is denoted $J^k(n, p)$. Let $\mathcal{G}_k$ be the normal subgroup of $\mathcal{G}$ consisting of those germs whose k-jet is equal to that of the identity. The standard k-jet group is defined to be the quotient group $\mathcal{G}/\mathcal{G}_k$ and is denoted $J^k\mathcal{G}$. This is a Lie group and acts on the affine space $J^k(n, p)$; see [20, section 7]. Let H^k denote the image of $\mathcal{M}_n^k\mathcal{E}(n, p)$ in $J^k(n, p)$, the vector subspace of $J^k(n, p)$ consisting of the homogeneous jets of degree k. A subgroup $\mathcal{H}$ of $\mathcal{G}$ is a *Lie subgroup* of the group $\mathcal{G}_1$ if for each k the jet group $J^k\mathcal{H}$ is a Lie subgroup of $J^k\mathcal{G}_1$.

Proposition 2.2. *Let $\mathcal{H}$ be a Lie subgroup of $\mathcal{G}_1$ and let f be a k-jet in the jet space $J^k(n, p)$. If $T \subset H^{k+1}$ is a vector subspace of H^{k+1} such that*

$$H^{k+1} \subset T + L(J^{k+1}\mathcal{H}) \cdot f$$

then every $(k+1)$-jet F with $j^k F = f$ is in the same $J^{k+1}\mathcal{H}$-orbit as some $(k+1)$-jet of the form $f + t$, for some $t \in T$.

Proof. The key observation to make is that for all $h \in H^{k+1}$ and $l \in L(J^{k+1}\mathcal{H})$ we have $l \cdot (f + h) = l \cdot f$, since $L(J^{k+1}\mathcal{H}) \subset L(J^{k+1}\mathcal{G}_1)$. Therefore condition (3) in section 1 holds when we take $A = J^{k+1}(n, p)$, $W = H^{k+1}$, $G = J^{k+1}\mathcal{H}$. We may consequently apply proposition 1.3 and deduce the result. □

We illustrate the method by applying it to some fairly simple examples.

Example 2.3. We consider $\mathcal{A}$-equivalence here, that is $\mathcal{H} = \mathcal{A}_1$ above.

(i) (See [24, 29].) Consider maps $f : \boldsymbol{R}^2, 0 \to \boldsymbol{R}^2, 0$ of corank 1. Clearly such an f has 1-jet equivalent to $(x, 0)$. Since $L(J^2\mathcal{A}_1).f$ contains $(\phi, 0)$, for any homogeneous ϕ of degree 2, and $(0, x^2)$ we deduce that $(x, a_1xy + a_2y^2)$, where $a_1, a_2 \in \boldsymbol{R}$, is a complete transversal. Depending on whether $a_2 \neq 0$; $a_2 = 0$, $a_1 \neq 0$; or $a_1 = a_2 = 0$ we arrive at three possible 2-jets (x, y^2), (x, xy), $(x, 0)$. The first is easily seen to be 2-$\mathcal{A}_1$ determined.

(ii) One can proceed to a complete classification of corank 1 germs in low codimension. The method often produces a natural series of singularities. Consider for example germs with 3-jet equivalent to (x, y^3). If we view this as a k-jet ($k \geqslant 3$) then a complete $(k+1)$-transversal is given by $(x, y^3 + ax^ky)$. If $a \neq 0$ then we reduce to the normal forms $(x, y^3 \pm x^ky)$ for k even and $(x, y^3 + x^ky)$ if k is odd. Since these germs are $(k+1)$-$\mathcal{A}$ determined we have a complete list of $\mathcal{A}$-finitely determined germs with the given 3-jet.

(iii) Finally consider maps $f : \boldsymbol{R}^2, 0 \to \boldsymbol{R}^3, 0$ with 2-jet $(x, y^2, 0)$. A complete transversal in the space of 3-jets is given by $(x, y^2, a_1x^2y + a_2y^3)$. A little linear algebra shows we can reduce to one of the jets $(x, y^2, x^2y \pm y^3)$, (x, y^2, x^2y), (x, y^2, y^3) or $(x, y^2, 0)$. The first of these jets is 3-$\mathcal{A}$ determined and need not be considered further. If we regard (x, y^2, x^2y) now as a k-jet ($k \geqslant 3$) a complete $(k+1)$-transversal is given by $(x, y^2, x^2y + ay^{k+1})$ if k is even and (x, y^2, x^2y) if k is odd. In the former case, if $a \neq 0$ we reduce to $(x, y^2, x^2y \pm y^{k+1})$ which is $(k+1)$-$\mathcal{A}$ determined. In the remaining cases the procedure can be repeated and this essentially completes the classification of $\mathcal{A}$-finitely determined germs with this 3-jet. When the k-jet is (x, y^2, y^3) a complete $(k+1)$-transversal is $(x, y^2, y^3 + ax^ky)$. If $a \neq 0$ then we reduce to $(x, y^2, y^3 \pm x^ky)$ which is again $(k+1)$-$\mathcal{A}$ determined and the classification is again complete. (This classification is due to Mond [22].)

Remark 2.4. (i) Having produced the complete transversal one can reduce the number of representatives even further by scaling in the source and target. The corresponding group consists of pairs of invertible diagonal matrices. In (i) above we also needed another type of transformation, to show that any jet of the form $(x, \alpha xy + y^2)$ is $\mathcal{A}$-equivalent to (x, y^2). Our aim is to minimize the need for such *ad hoc* calculations.

(ii) Useful as proposition 2.2 is there are a number of ways in which one would like it improved. The condition that the group $\mathcal{H}$ above is a subgroup of $\mathcal{G}_1$ implies in particular that the jet groups $J^k\mathcal{H}$ are unipotent. This was the key to the determinacy results obtained in [9]. It is consequently natural to seek a result such as proposition 2.2 above, which is valid for a 'unipotent' $\mathcal{H}$, and indeed we shall establish a useful complete transversal result for such larger subgroups of $\mathcal{A}$ and $\mathcal{K}$ using the filtration introduced in [9]. The choice of subgroups and associated filtration are intimately connected. We discuss the general problem of unipotent actions in [7].

2.2. Unipotent groups

We recall the following from [9]. Let $\mathcal{G}$ be a subgroup of $\mathcal{K}$. We call $\mathcal{G}$ *jet closed* if for each $r \geqslant 1$, $J^r\mathcal{G}$ is a closed subgroup of $J^r\mathcal{K}$. If $\mathcal{G}$ is jet closed it follows that $J^s(L\mathcal{G}) \subset L(J^s\mathcal{G})$ for all s. In many cases we have equality. If a jet closed group $\mathcal{G}$ satisfies $J^s(L\mathcal{G}) = L(J^s\mathcal{G})$ for all s then we call it *fibrant*. We find that $\mathcal{R}$, $\mathcal{L}$, $\mathcal{A}$, $\mathcal{C}$ and $\mathcal{K}$ are all jet closed and fibrant. Further examples are given via the following concept. Let $\mathcal{H}$ be a subgroup of $\mathcal{G}$, then $\mathcal{H}$ is said to be *strongly closed* in $\mathcal{G}$ if $\mathcal{H}_s = \mathcal{G}_s$ for some s (equivalently $\mathcal{G}_s \subset \mathcal{H}$), and $J^s\mathcal{H}$ is closed in $J^s\mathcal{G}$. Now, a strongly closed subgroup $\mathcal{H}$ of a jet closed group $\mathcal{G}$ is itself jet closed. If, in addition, $\mathcal{G}$ is fibrant then so is $\mathcal{H}$. See [9, section 4] for more details.

In this section we will be concerned mainly with fibrant subgroups $\mathcal{G}$ of $\mathcal{K}$ such that $L(J^1\mathcal{G})$ acts nilpotently on $\boldsymbol{K}^{n+p}$. The following was noted in [9].

Proposition 2.5. *Let $\mathcal{G}$ be a subgroup of $\mathcal{K}$ such that $L(J^1\mathcal{G})$ acts nilpotently on $\boldsymbol{K}^{n+p}$. Then given an integer $r \geqslant 1$ there is an integer $k_r \geqslant 1$ such that*

$$(L\mathcal{G})^{k_r} \cdot (\mathcal{M}_n^r\mathcal{E}(n, p)) \subset \mathcal{M}_n^{r+1}\mathcal{E}(n, p).$$

Definition 2.6. For such $\mathcal{G}$ and for integers $r \geqslant 1$ and $s \geqslant 0$ we define the *nilpotent filtration*

$$M_{r,s}(\mathcal{G}) = \sum_{i \geqslant s} (L\mathcal{G})^i \cdot (\mathcal{M}_n^r\mathcal{E}(n, p)) + \mathcal{M}_n^{r+1}\mathcal{E}(n, p).$$

For $r = 0$ we just define $M_{0,0}(\mathcal{G})$ to be $\mathcal{M}_n\mathcal{E}(n, p)$ for consistency. The associated (r, s)-*jet space* $J^{r,s}(n, p)$ is then defined to be $\mathcal{M}_n\mathcal{E}(n, p)/M_{r,s}(\mathcal{G})$, and by the *homogeneous* terms of degree (r, s) we mean the image of the space $M_{r,s-1}(\mathcal{G})$ in this quotient, and denote this $H^{r,s}$. Given $f \in \mathcal{M}_n\mathcal{E}(n, p)$ we denote its image in $J^{r,s}(n, p)$ by $j^{r,s}f$. We filter the group $\mathcal{G}$ using the normal subgroups $\mathcal{G}_k$ (see definition 2.1); the associated r-*jet group* $J^r\mathcal{G}$ is then defined to be $\mathcal{G}/\mathcal{G}_r$. For the subgroups $\mathcal{G}$ of $\mathcal{K}$ which are used in practice it is already known that $J^r\mathcal{G}$ is a Lie group. We will assume this holds for $\mathcal{G}$ in what follows.

Remark 2.7. Note that by proposition 2.5 the sum used to define $M_{r,s}(\mathcal{G})$ is finite. It also follows that for each r there is an integer k_r such that

$$M_{r,k_r} = M_{r,k_r+1} = M_{r,k_r+2} = \cdots = \mathcal{M}_n^{r+1}\mathcal{E}(n, p) = M_{r+1,0}.$$

So this filtration is *not* doubly indexed as might be thought at first, but can be thought of as a singly indexed sequence. The definition of $H^{r,s}$ as the image of $M_{r,s-1}(\mathcal{G})$ in $J^{r,s}(n, p)$ makes sense for $s = 0$ as $M_{r,0} = M_{r-1,k_{(r-1)}}$.

We recall the following, [9, lemma 2.3].

Lemma 2.8. *Let $f \in \mathcal{M}_n\mathcal{E}(n.p)$, $b \in M_{r,s}(\mathcal{G})$ and $l \in L\mathcal{G}$. Then*

$$l \cdot (f + b) - l \cdot f \in M_{r,s+1}(\mathcal{G}).$$

The required complete transversal result is the following:

Theorem 2.9. *Let $\mathcal{G}$ be a fibrant subgroup of $\mathcal{K}$ such that $L(J^1\mathcal{G})$ acts nilpotently on $\boldsymbol{K}^{n+p}$. Let f be a smooth germ $\boldsymbol{K}^n, 0 \to \boldsymbol{K}^p, 0$ and let T be a subspace of $M_{r,s}(\mathcal{G})$ with*

$$M_{r,s}(\mathcal{G}) \subset T + L\mathcal{G} \cdot f + M_{r,s+1}(\mathcal{G}).$$

Then any germ $g : \boldsymbol{K}^n, 0 \to \boldsymbol{K}^p, 0$ with $g - f \in M_{r,s}(\mathcal{G})$ is $\mathcal{G}$ equivalent to a germ of the form $f + t + \phi$ with $t \in T$ and $\phi \in M_{r,s+1}(\mathcal{G})$.

Proof. Since $M_{r,s}(\mathcal{G})$ contains $\mathcal{M}_n^{r+1}\mathcal{E}(n,p)$ it follows that it is of finite codimension in $\mathcal{M}_n\mathcal{E}(n,p)$ and $J^{r,s}(n,p)$ is, therefore, a finite dimensional vector space. From [20, section 7] the standard r-jet group $J^r\mathcal{G}$ is a Lie group and acts on the standard r-jet space $\mathcal{M}_n\mathcal{E}(n,p)/\mathcal{M}_n^{r+1}\mathcal{E}(n,p)$, which we will denote by U. Let V denote the image of $M_{r,s+1}(\mathcal{G})$ under the natural projection into U, let $j^r f$ denote the image of $f \in \mathcal{M}_n\mathcal{E}(n,p)$ in U, $j^r v$ the image of $v \in M_{r,s+1}(\mathcal{G})$ in V and $j^r l$ the image of $l \in L\mathcal{G}$ in $J^r(L\mathcal{G})$. (Since $\mathcal{G}$ is fibrant we have $J^r(L\mathcal{G}) = L(J^r\mathcal{G})$, so any element of $L(J^r\mathcal{G})$ may be written as $j^r l$ for some $l \in L\mathcal{G}$.) Then

$$j^r l \cdot (j^r f + j^r v) - j^r l \cdot (j^r f) = j^r(l \cdot (f+v) - l \cdot f) \in V$$

by lemma 2.8. Thus, by lemma 1.2, the action of $J^r\mathcal{G}$ on U induces an action on U/V. But $\mathcal{M}_n^{r+1}\mathcal{E}(n,p) \subset M_{r,s+1}(\mathcal{G}) \subset \mathcal{M}_n\mathcal{E}(n,p)$ and

$$V = M_{r,s+1}(\mathcal{G})/\mathcal{M}_n^{r+1}\mathcal{E}(n,p)$$

therefore

$$U/V \cong \mathcal{M}_n\mathcal{E}(n,p)/M_{r,s+1}(\mathcal{G}).$$

We conclude that the action of $J^r\mathcal{G}$ on $\mathcal{M}_n\mathcal{E}(n,p)/\mathcal{M}_n^{r+1}\mathcal{E}(n,p)$ induces an action on $J^{r,s+1}(n,p)$.

By assumption $g = f + \tilde{f}$ for some $\tilde{f} \in M_{r,s}(\mathcal{G})$, that is $j^{r,s+1}g = j^{r,s+1}f + j^{r,s+1}\tilde{f}$ where $j^{r,s+1}\tilde{f} \in H^{r,s+1}$. (Recall that $H^{r,s+1}$ is the image of $M_{r,s}(\mathcal{G})$ in $J^{r,s+1}(n,p)$.) We now apply the proposition 1.3 taking $G = J^r\mathcal{G}$, $A = J^{r,s+1}(n,p)$ and $W = H^{r,s+1}$. Condition (3) of the proposition holds because given $j^r l \in L(J^r\mathcal{G})$ and $j^{r,s+1}h \in H^{r,s+1}$ (that is $h \in M_{r,s}(\mathcal{G})$) we have

$$j^r l \cdot (j^{r,s+1}f + j^{r,s+1}h) - j^r l \cdot (j^{r,s+1}f) = j^{r,s+1}(l \cdot (f+h) - l \cdot f) = 0$$

using lemma 2.8 again. The result now follows. □

To implement the method one needs a more explicit identification of the nilpotent filtration. One way to do this is to order the degree k monomial vectors using a system of weights. In what follows $\alpha = (\alpha_1, \ldots, \alpha_n)$ and $\beta = (\beta_1, \ldots, \beta_p)$ will denote the source and target weights respectively. The monomial vector $x_1^{k_1} \ldots x_n^{k_n} e_i$ is assigned a weight $k_1\alpha_1 + \cdots + k_n\alpha_n - \beta_i$. The $\mathcal{E}_n$-submodule of $\mathcal{M}_n\mathcal{E}(n,p)$ generated by such monomial vectors of weight $\geqslant k$ is denoted $F^k_{\alpha,\beta}\mathcal{E}(n,p)$. Weighted filtrations will be discussed fully in section 2.3. We use weights here only as a means of identifying the nilpotent filtration. In particular, we do not impose the same restrictions on α and β as in section 2.3.

We consider the case of $\mathcal{A}$-classification, though the method extends to other subgroups of $\mathcal{K}$. Let $(x_1, \ldots, x_n)$ denote coordinates on $(\boldsymbol{K}^n, 0)$ and $(y_1, \ldots, y_p)$ those on $(\boldsymbol{K}^p, 0)$. Let $\mathcal{G}$ be a subgroup of $\mathcal{A}$ such that $L(J^1\mathcal{G})$ acts nilpotently on $\boldsymbol{K}^{n+p}$. For 'large enough $\mathcal{G}$' (we make this precise below) we can assign source and target weights such that the partition of the monomial vectors of (standard) degree k via their weight corresponds to their partition into the (k,s)-jet levels using the nilpotent filtration.

Proposition 2.10. *Suppose $L\mathcal{G}$ contains the following vectors and assign source and target weights according to the case in question.*

Vectors	*Weight*
$x_i\partial/\partial x_{i+1} \in L\mathcal{R}$ *or*	$\alpha = (n, \ldots, 2, 1)$
$x_{i+1}\partial/\partial x_i \in L\mathcal{R}$	$\alpha = (1, 2, \ldots, n)$
$y_j\partial/\partial y_{j+1} \in L\mathcal{L}$ *or*	$\beta = (0, -1, \ldots, -p+1)$
$y_{j+1}\partial/\partial y_j \in L\mathcal{L}$	$\beta = (-p+1, \ldots, -1, 0)$

for $i = 1, \ldots, n-1$ and $j = 1, \ldots, p-1$. Then

$$\sum_{i \geqslant s} (L\mathcal{G})^i \cdot (\mathcal{M}_n^k \mathcal{E}(n, p)) + \mathcal{M}_n^{k+1} \mathcal{E}(n, p) = (F_{\alpha,\beta}^{k+s} \mathcal{E}(n, p) \cap \mathcal{M}_n^k \mathcal{E}(n, p)) + \mathcal{M}_n^{k+1} \mathcal{E}(n, p).$$

So for fixed k, the $M_{k,s}(\mathcal{G})$ filtration can be replaced by the weighted filtration modulo $\mathcal{M}_n^{k+1}\mathcal{E}(n, p)$, that is the filtration on the right-hand side of the above expression. Recall that the space of 'homogeneous terms' of degree (k, s), $H^{k,s}$, is the image of $M_{k,s-1}(\mathcal{G})$ in the jet space $\mathcal{M}_n\mathcal{E}(n, p)/M_{k,s}(\mathcal{G})$. In particular, the homogeneous monomial vectors of degree (k, s) are just those of (standard) degree k with weight $k + s - 1$.

Proof. We shall consider the second and third combinations in the table, that is where $\alpha = (1, 2, \ldots, n)$ and $\beta = (0, -1, \ldots, -p+1)$. Observe that with the choice of weights, $\mathcal{M}_n^r \mathcal{E}(n, p) \subset F_{\alpha,\beta}^r \mathcal{E}(n, p)$ for any $r \geqslant 1$, also that the action of $L\mathcal{G}$ increases weight, that is $L\mathcal{G} \cdot (F_{\alpha,\beta}^r \mathcal{E}(n, p)) \subset F_{\alpha,\beta}^{r+1} \mathcal{E}(n, p)$. Thus

$$\sum_{i \geqslant s} (L\mathcal{G})^i \cdot (\mathcal{M}_n^k \mathcal{E}(n, p)) \subset F_{\alpha,\beta}^{k+s} \mathcal{E}(n, p) \cap \mathcal{M}_n^k \mathcal{E}(n, p).$$

For the reverse inclusion we need only consider monomial vectors of degree k in $F_{\alpha,\beta}^{k+s} \mathcal{E}(n, p) \cap \mathcal{M}_n^k \mathcal{E}(n, p)$; let $x_1^{k_1} \ldots x_n^{k_n} e_j$ be such a vector. Thus

$$k_1 + 2k_2 + 3k_3 + \cdots + nk_n \geqslant k + s + (1 - j). \tag{4}$$

Now

$$\begin{aligned}
&(x_2 \partial/\partial x_1)^{k-k_1} \cdot (x_1^k) = x_1^{k_1} x_2^{k-k_1} \\
&(x_3 \partial/\partial x_2)^{k-k_1-k_2} \cdot (x_1^{k_1} x_2^{k-k_1}) = x_1^{k_1} x_2^{k_2} x_3^{k-k_1-k_2} \\
&\vdots
\end{aligned}$$

and continuing in this fashion up to

$$\begin{aligned}
&(x_n \partial/\partial x_{n-1})^{k-k_1-k_2-\cdots-k_{n-1}} \cdot (x_1^{k_1} x_2^{k_2} \ldots x_{n-2}^{k_{n-2}} x_{n-1}^{k-k_1-k_2-\cdots-k_{n-2}}) \\
&\qquad = x_1^{k_1} x_2^{k_2} \ldots x_{n-2}^{k_{n-2}} x_{n-1}^{k_{n-1}} x_n^{k-k_1-k_2-\cdots-k_{n-1}}.
\end{aligned}$$

But $k = k_1 + \cdots + k_n$ and $x_{i+1}\partial/\partial x_i \in L\mathcal{G}$ by assumption, so for $1 \leqslant l \leqslant p$

$$x_1^{k_1} \ldots x_n^{k_n} e_l \in (L\mathcal{G})^r \cdot (\mathcal{M}_n^k \mathcal{E}(n, p))$$

where

$$\begin{aligned}
r &= (n-1)k - (n-1)k_1 - (n-2)k_2 - (n-3)k_3 - \cdots - k_{n-1} \\
&= k_2 + 2k_3 + 3k_4 + \cdots + (n-2)k_{n-1} + (n-1)k_n.
\end{aligned}$$

In particular, taking $l = 1$ and applying the $j - 1$ vector fields

$$y_1 \partial/\partial y_2 \qquad y_2 \partial/\partial y_3 \qquad \ldots \qquad y_{j-1} \partial/\partial y_j$$

in sequence to $x_1^{k_1} \ldots x_n^{k_n} e_1$ shows that $x_1^{k_1} \ldots x_n^{k_n} e_j \in (L\mathcal{G})^{r+j-1} \cdot (\mathcal{M}_n^k \mathcal{E}(n, p))$. But from relation (4) we deduce

$$k_2 + 2k_3 + 3k_4 + \cdots + (n-1)k_n + j - 1 \geqslant s$$

that is $r + j - 1 \geqslant s$ and

$$x_1^{k_1} \ldots x_n^{k_n} e_j \in \sum_{i \geqslant s} (L\mathcal{G})^i \cdot (\mathcal{M}_n^k \mathcal{E}(n, p)).$$

□

The vectors referred to in proposition 2.10 are the 'extra' vectors present in $L\mathcal{A} \setminus L\mathcal{A}_1$. For classification purposes one would prefer to use some unipotent group $\mathcal{G}$ such that the nilpotent Lie algebra $L\mathcal{G} \subset L\mathcal{A}$ contains as many of these 'extra' vectors as possible. There are four natural cases to consider:

$$L\mathcal{G} = L\mathcal{A}_1 \oplus \boldsymbol{K}\{x_i\partial/\partial x_j\} \oplus \boldsymbol{K}\{y_k\partial/\partial y_l\}$$

for all $i < j$ (or alternatively all $i > j$) and similarly for k and l. Such cases are used in practical applications and proposition 2.10 applies.

Example 2.11. The following examples should clarify the theory.

(i) Let us reconsider example 2.3(i), that is map-germs $\boldsymbol{R}^2, 0 \to \boldsymbol{R}^2, 0$ of corank 1. Let (x, y) denote coordinates in the source and (u, v) those in the target. This time we extend the group $\mathcal{A}_1$ and work with the unipotent subgroup of $\mathcal{A}$ having nilpotent Lie algebra

$$L\mathcal{A}_1 \oplus \boldsymbol{R}\{x\partial/\partial y\} \oplus \boldsymbol{R}\{v\partial/\partial u\}.$$

Each (r, s)-jet space is just a refinement of the standard r-jet space (by degree) and a convenient way to describe these spaces is to list the generators for each of the spaces $H^{r,s}$; see table 1. The 'weight' column demonstrates the use of weights to partition the monomial vectors into their (r, s)-levels, as discussed in proposition 2.10. Here $\alpha = (2, 1)$ and $\beta = (-1, 0)$.

Table 1. Generators for the spaces $H^{r,s}$.

(r, s)	Basis for $H^{r,s}$	Weight
$(1, 0)$	$\{0\}$	
$(1, 1)$	$\{(0, y)\}$	1
$(1, 2)$	$\{(y, 0), (0, x)\}$	2
$(1, 3)$ or $(2, 0)$	$\{(x, 0)\}$	3
$(2, 1)$	$\{(0, y^2)\}$	2
$(2, 2)$	$\{(y^2, 0), (0, xy)\}$	3
$(2, 3)$	$\{(xy, 0), (0, x^2)\}$	4
$(2, 4)$ or $(3, 0)$	$\{(x^2, 0)\}$	5

Now consider the 1-jet $(x, 0)$. A $(2, 1)$-transversal is $\{(0, y^2)\}$ and we obtain jets of the form (x, ay^2) for $a \in \boldsymbol{R}$. Apply scaling coordinate changes to reduce this to the two cases (x, y^2) and $(x, 0)$. In the first case all of the higher $(2, s)$-transversals are found to be empty; while in the second case the only non-empty transversal is the $(2, 2)$-transversal: $\{(0, xy)\}$. After further scaling we arrive at the complete list of 2-jets over $(x, 0)$, namely (x, y^2), (x, xy) and $(x, 0)$ as before. Though a simple example, this demonstrates the streamlining we are achieving in our classification techniques. Now to a more substantial example.

(ii) (See [16].) Let us consider map-germs $\boldsymbol{R}^3, 0 \to \boldsymbol{R}^4, 0$ of corank 1. Let (x, y, z) denote coordinates in the source and (u_1, u_2, u_3, u_4) those in the target. The 1-jet of such a map-germ is $\mathcal{A}$-equivalent to $(x, y, 0, 0)$ and with a little work similar to that which follows we obtain the following complete list of 2-jets:

$$(x, y, z^2, xz) \qquad (x, y, z^2, 0) \qquad (x, y, yz, xz) \qquad (x, y, xz, 0) \qquad (x, y, 0, 0).$$

We will consider the third as our example. Consider the unipotent subgroup of $\mathcal{A}$ having nilpotent Lie algebra

$$L\mathcal{A}_1 \oplus \boldsymbol{R}\{x\partial/\partial y, x\partial/\partial z, y\partial/\partial z\} \oplus \boldsymbol{R}\{u_i\partial/\partial u_j \text{ for } i > j\}.$$

Table 2. Generators for the spaces $H^{3,s}$.

$(3, s)$	Basis for $H^{3,s}$	Weight
$(3, 1)$	$\{(0, 0, 0, z^3)\}$	3
$(3, 2)$	$\{(0, 0, z^3, 0), (0, 0, 0, yz^2)\}$	4
$(3, 3)$	$\{(0, z^3, 0, 0), (0, 0, yz^2, 0), (0, 0, 0, y^2z), (0, 0, 0, xz^2)\}$	5
$(3, s \geqslant 4)$	etc ...	

The spaces $H^{3,s}$ are somewhat tedious to write down, though only the first few levels feature in the calculations. These are shown in table 2 (the required weights being $\alpha = (3, 2, 1)$ and $\beta = (-3, -2, -1, 0)$).

Put $f = (x, y, yz, xz)$; a $(3, 1)$-transversal gives jets over f to be of the form $(x, y, yz, xz + az^3)$, $a \in \boldsymbol{R}$. Let us apply scaling coordinate changes to simplify matters before proceeding. We obtain the jets $(x, y, yz, xz + z^3)$ and (x, y, yz, xz). In both cases the only higher non-empty $(3, s)$-transversal is the $(3, 2)$-transversal and we obtain the 3-jets $(x, y, yz + bz^3, xz + z^3)$ and $(x, y, yz + bz^3, xz)$, for some $b \in \boldsymbol{R}$. In the first case it is tempting to scale b to 0 or 1. However, the resulting 3-jets have the same $J^3\mathcal{A}$ codimension, indeed the codimension is found to be constant for all b, suggesting that the family is $\mathcal{A}$ trivial. A simple application of the Mather lemma (performed using the computer package written by the second author) shows that this is so. We, therefore, have the one normal form: $(x, y, yz, xz+z^3)$. This demonstrates that we cannot always avoid redundancy in the classification procedure even by extending the group $\mathcal{A}_1$ to unipotent subgroups of $\mathcal{A}$. (We have extended the Lie algebra $L\mathcal{A}_1$ by the maximum number of generators.) Returning to the second jet $(x, y, yz + bz^3, xz)$; here we apply scaling coordinate changes and reduce this to the 3-jets $(x, y, yz + z^3, xz)$ (equivalent to $(x, y, yz, xz + z^3)$ obtained earlier) and (x, y, yz, xz). The complete list of 3-jets over (x, y, yz, xz) is therefore

$$(x, y, yz, xz + z^3) \qquad (x, y, yz, xz).$$

If instead we use the group $\mathcal{A}_1$, a 3-transversal over (x, y, yz, xz) is

$$(x, y, yz + a_1z^3 + a_2xz^2, xz + a_3z^3 + a_4xz^2 + a_5yz^2) \qquad \text{for } a_i \in \boldsymbol{R}.$$

Of course, one may reduce these to the two cases above, but this would involve a lot of (*ad hoc*) work.

Remark 2.12. In applications we are normally interested in subgroups of $\mathcal{K}$ which are not unipotent; let us take $\mathcal{A}$ as an example. We apply the above theory taking some 'large enough' unipotent subgroup of $\mathcal{A}$ and this provides an efficient classification scheme. A natural question is, can the complete transversal theorem be extended to cover non-unipotent subgroups, in particular $\mathcal{A}$ itself? The answer, in general, is no. For suppose that proposition 2.2 holds for $\mathcal{H} = \mathcal{A}$ (ignoring the reference to $\mathcal{G}$ given there). Then following similar arguments to [9] theorems 1.9 and 2.1 we arrive at the false determinacy result: if $\mathcal{M}_n^{k+1}\mathcal{E}(n, p) \subset L\mathcal{A} \cdot f$ then f is k-$\mathcal{A}$ determined. However, there are well known counter-examples to this. (Indeed, in the case $\mathcal{H} = \mathcal{R}$ the result states that the function-germ x^{k+1} is k-$\mathcal{R}$ determined!) Thus, in a sense, our unipotent classification scheme is the best one can hope for.

2.3. Weighted homogeneous filtrations

Although the earlier results of this section provide useful techniques, some classifications lend themselves more naturally to the use of weighted filtrations. In addition, the methods

discussed below provide further improvement on the basic complete transversal result, proposition 2.2. The original ideas were introduced by Arnold for the $\mathcal{R}$ case, [3]. These extend to the other subgroups of $\mathcal{K}$, a concise treatment is given in [9, section 5]. We use these as our two basic references and omit most of the proofs.

Definition 2.13. Let $\alpha = (\alpha_1, \ldots, \alpha_n)$ be a given sequence of positive integers referred to as *weights*. A function-germ $f : \boldsymbol{K}^n, 0 \to \boldsymbol{K}, 0$ is said to be *weighted homogeneous of weight* r (with respect to α) if

$$f(t^{\alpha_1}x_1, \ldots, t^{\alpha_n}x_n) = t^r f(x_1, \ldots, x_n)$$

for all $t \in \boldsymbol{K}$. For a monomial function $x_1^{k_1} \ldots x_n^{k_n}$ an equivalent condition is that $\alpha_1 k_1 + \cdots + \alpha_n k_n = r$. A polynomial function is weighted homogeneous if and only if all its constituent monomials are.

Alternatively one might allow rational weights and can then assume a weighted homogeneous polynomial is of weight 1.

Definition 2.14. The ideal in $\mathcal{M}_n\mathcal{E}_n$ generated by the monomials of weight r and greater is denoted $F^r_\alpha\mathcal{E}_n$, or just $F^r\mathcal{E}_n$ for short. The sequence of ideals $\{F^r_\alpha\mathcal{E}_n\}$, for $r \geqslant 1$, defines a filtration of the ring $\mathcal{M}_n\mathcal{E}_n$.

Note that $F^1_\alpha\mathcal{E}_n = \mathcal{M}_n$ (for all α). Also, when dealing with function-germs and $\mathcal{R}$ equivalence one sometimes works with $\mathcal{E}_n$ and the filtration $\{F^r_\alpha\mathcal{E}_n\}$ for $r \geqslant 0$.

For germs $f : \boldsymbol{K}^n, 0 \to \boldsymbol{K}^p, 0$ with $p > 1$ we must introduce weights for each target coordinate too.

Definition 2.15. For α as above, let $\beta = (\beta_1, \ldots, \beta_p)$ be a sequence of non-negative integers, and let $f = (f_1, \ldots, f_p)$ be a map-germ $\boldsymbol{K}^n, 0 \to \boldsymbol{K}^p, 0$. Then f is said to be *weighted homogeneous of weight* r (with respect to (α, β)) if f_i is weighted homogeneous of weight $r + \beta_i$ (as in definition 2.13), for $i = 1, \ldots, p$.

Definition 2.16. The submodule of $\mathcal{M}_n\mathcal{E}(n, p)$ generated by map-germs $f = (f_1, \ldots, f_p)$ with $f_i \in F^{r+\beta_i}_\alpha\mathcal{E}_n$, for $1 \leqslant i \leqslant p$, is denoted $F^r_{\alpha,\beta}\mathcal{E}(n, p)$. We will follow [9] and adopt the more compact notation $F^r_{\alpha,\beta}M$ when there is no risk of confusion. The sequence of submodules $\{F^r_{\alpha,\beta}M\}$, for $r \geqslant 0$, defines a filtration of the module $F^0_{\alpha,\beta}M$. (Note that generally $F^0_{\alpha,\beta}M \neq \mathcal{M}_n\mathcal{E}(n, p)$.)

We recall the following results from [9, section 5].

Lemma 2.17. *(i) The germ $f \in F^r_{\alpha,\beta}\mathcal{E}(n, p)$ if and only if, for all k and for all $h \in F^k_\beta\mathcal{E}_p$, $h \circ f \in F^{r+k}_\alpha\mathcal{E}_n$.*

(ii) If $f, g \in F^0_{\alpha,\beta}\mathcal{E}(n, p)$ with $(f - g) \in F^r_{\alpha,\beta}\mathcal{E}(n, p)$ and $h \in F^s_{\beta,\gamma}\mathcal{E}(p, q)$, then $(h \circ f - h \circ g) \in F^{r+s}_{\alpha,\gamma}\mathcal{E}(n, q)$.

In this section we will take $\mathcal{G}$ to be one of the standard Mather groups $\mathcal{R}$, $\mathcal{L}$, $\mathcal{A}$, $\mathcal{C}$ or $\mathcal{K}$, although the methods extend to other subgroups of $\mathcal{K}$. We begin by defining a filtration of such groups compatible with the weighted filtration of $\mathcal{M}_n\mathcal{E}(n, p)$; not all elements of $\mathcal{K}$ will respect the weighted filtration.

Definition 2.18. For $r \geqslant 0$ define

$$F^r\mathcal{R} = \{\phi \in \mathcal{R} : f \circ \phi - f \in F^{r+t}\mathcal{E}_n \text{ for all } f \in F^t\mathcal{E}_n \text{ and all } t\}.$$

(Arnold calls this the *group of diffeomorphisms of filtration* r.)

This can be put in more concrete terms.

Proposition 2.19. *The group $F^r\mathcal{R}$ can be rewritten*

$$(1_n + F^r_{\alpha,\alpha}\mathcal{E}(n,n)) \cap \mathcal{R}.$$

One can similarly set $F^r\mathcal{L} = (1_p + F^r_{\beta,\beta}\mathcal{E}(p,p)) \cap \mathcal{L}$. Finally consider the group $\mathcal{C}$ consisting of diffeomorphisms $\boldsymbol{K}^{n+p}, 0 \to \boldsymbol{K}^{n+p}, 0$ which project to the identity on $\boldsymbol{K}^n$ and leave fixed the subspace $\boldsymbol{K}^n \times \{0\}$. So

$$\mathcal{C} = 1_{n+p} + \mathcal{M}_p\mathcal{E}(n+p,p)$$

where we consider $\mathcal{E}(n+p,p)$ as a subspace of $\mathcal{E}(n+p,n+p)$ (by associating $f \in \mathcal{E}(n+p,p)$ with the germ $(x,y) \mapsto (0, f(x,y))$ in $\mathcal{E}(n+p,n+p)$) and consider $\mathcal{M}_p$ as an ideal of $\mathcal{E}_{n+p}$ (via the induced map $\pi^* : \mathcal{E}_p \to \mathcal{E}_{n+p}$ determined by the natural projection π). We define $F^r\mathcal{C}$ analogously to the $\mathcal{R}$ case.

Definition 2.20. For $r \geqslant 0$ define

$$F^r\mathcal{C} = (1_{n+p} + F^r_{\alpha\cup\beta,\alpha\cup\beta}\mathcal{E}(n+p,n+p)) \cap \mathcal{C}$$

where $\alpha \cup \beta$ denotes the sequence $(\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_p)$. Note that if $\mathcal{D}_{n+p}$ denotes the group of diffeomorphisms of $\boldsymbol{K}^{n+p}$ at 0 then

$$F^r\mathcal{C} = F^r\mathcal{D}_{n+p} \cap \mathcal{C}.$$

We will need a characterization for $F^r\mathcal{C}$ analogous to that of $F^r\mathcal{R}$ in proposition 2.19.

Proposition 2.21. *(i) If $f, g \in F^0_{\alpha,\beta}M$ with $(f-g) \in F^t_{\alpha,\beta}M$ and $\phi \in F^r\mathcal{C}$, then the action $(\cdot)$ of $F^r\mathcal{C}$ on $F^0_{\alpha,\beta}M$ defined by $(1_n, \phi\cdot f) = \phi \circ (1_n, f)$ satisfies*

$$(\phi\cdot f - f) - (\phi\cdot g - g) \in F^{r+t}_{\alpha,\beta}M.$$

(ii) We have

$$F^r\mathcal{C} = \{\phi \in \mathcal{C} : \phi\cdot f - f \in F^{r+t}_{\alpha,\beta}M \text{ for all } f \in F^t_{\alpha,\beta}M \text{ and all } t\}.$$

It is now an easy matter to establish the following two lemmas. The importance of 'weighted filtrations' comes from the fact that the subgroups $F^r\mathcal{G}$ respect the filtration and there exists an induced action at the jet-level.

Lemma 2.22. *(i) Each subgroup $F^r\mathcal{G}$ respects the filtration $\{F^r_{\alpha,\beta}M\}$, that is for $\phi \in F^r\mathcal{G}$ and $f \in F^s_{\alpha,\beta}M$ we have $\phi\cdot f \in F^s_{\alpha,\beta}M$.*

(ii) Each $F^r\mathcal{G}$ is a normal subgroup of $F^0\mathcal{G}$.

(iii) For $s, l, r \geqslant 0$ the action of $F^r\mathcal{G}$ on $\mathcal{M}_n\mathcal{E}(n,p)$ induces an action on $F^s_{\alpha,\beta}M/F^{s+l}_{\alpha,\beta}M$.

(iv) In addition, if $m \geqslant 0$ and $l \leqslant r+m$ then the action in (iii) induces an action of $F^r\mathcal{G}/F^{r+m}\mathcal{G}$ on $F^s_{\alpha,\beta}M/F^{s+l}_{\alpha,\beta}M$.

Corollary 2.23. *In particular, for $s = 0$, $l = k+1$, $r \geqslant 0$ and $m = k+1-r \geqslant 0$, there is an action of the quotient group (Lie group, in fact) $F^r\mathcal{G}/F^{k+1}\mathcal{G}$ on $F^0_{\alpha,\beta}M/F^{k+1}_{\alpha,\beta}\mathcal{G}$.*

The complete transversal theorem we establish requires $r > 0$, the most significant case $r = 1$ yielding the group $F^1\mathcal{G}/F^{k+1}\mathcal{G}$. This is analogous to the $\mathcal{G}_1$ complete transversal result, proposition 2.2, where the standard filtration by degree is used. Here we are using a weighted filtration and a possibly larger group, $F^1\mathcal{G}$. It was noted in [9] that the groups $F^r\mathcal{G}$ are strongly closed subgroups of $\mathcal{G}$; this is useful for applications to determinacy. It also follows that each group is jet closed, a property which is needed for the complete transversal theorems.

Lemma 2.24. *(i) The Lie algebra of $F^r\mathcal{R}$ can be identified with $F^r_{\alpha,\alpha}\mathcal{E}(n,n)$; and that of $F^r\mathcal{L}$ with $F^r_{\beta,\beta}\mathcal{E}(p,p)$.*

(ii) The Lie algebra of $F^r\mathcal{C}$ can be identified with $F^r_{\alpha\cup\beta,\beta}\mathcal{E}(n+p,p)$.

(iii) The Lie algebra action satisfies the following approximation lemma. If $f, g \in F^0_{\alpha,\beta}\mathcal{E}(n,p)$ with $f-g \in F^t_{\alpha,\beta}\mathcal{E}(n,p)$ and $l \in L(F^r\mathcal{G})$ then

$$l \cdot f - l \cdot g \in F^{r+t}_{\alpha,\beta}\mathcal{E}(n,p).$$

From lemma 2.24(iii) it follows that the action of the Lie algebra of these groups respects the filtration, in that for $f \in F^t_{\alpha,\beta}M$ and $l \in L(F^r\mathcal{G})$ we have $l \cdot f \in F^{r+t}_{\alpha,\beta}M$. We say l is *weighted homogeneous* of weight r.

Theorem 2.25. *Let $\mathcal{G}$ be one of the standard Mather groups $\mathcal{R}$, $\mathcal{L}$, $\mathcal{A}$, $\mathcal{C}$ or $\mathcal{K}$ and $\alpha = (\alpha_1, \ldots, \alpha_n)$, $\beta = (\beta_1, \ldots, \beta_p)$ be sequences of weights as above. If $f \in F^0_{\alpha,\beta}M$, $k \geqslant 1$ and T is a subspace of $F^{k+1}_{\alpha,\beta}M$ such that*

$$F^{k+1}_{\alpha,\beta}M \subset T + L(F^1\mathcal{G}) \cdot f + F^{k+2}_{\alpha,\beta}M$$

then any germ g with $g - f \in F^{k+1}_{\alpha,\beta}M$ is $F^1\mathcal{G}$-equivalent to $f + t + \phi$ for some $t \in T$ and $\phi \in F^{k+2}M$.

Proof. The set $J^{k+1}(n,p) = F^0_{\alpha,\beta}M/F^{k+2}_{\alpha,\beta}M$ is a finite dimensional vector space since $F^{k+2}_{\alpha,\beta}M$ contains $\mathcal{M}^N_n.\mathcal{E}(n,p)$ for large enough N. Similarly $J^{k+1}\mathcal{G} = F^1\mathcal{G}/F^{k+2}\mathcal{G}$ is a Lie group of weighted homogeneous diffeomorphisms truncated by weight. That $J^{k+1}\mathcal{G}$ acts on $J^{k+1}(n,p)$ follows from corollary 2.23, and this action is smooth, being described locally by the composition of polynomial map-germs truncated by weight. Since $\mathcal{G}$ is jet closed (see section 2.2 and [9]) we have $J^s(L\mathcal{G}) \subset L(J^s\mathcal{G})$ for all s. Define L to be the Lie algebra $L(F^1\mathcal{G}) \subset L\mathcal{G}$. Now for all s, $J^sL \subset L(J^s\mathcal{G})$ is a Lie subalgebra, and for $f \in F^0_{\alpha,\beta}M$, $h \in F^s_{\alpha,\beta}M$ and $l \in L$ we have

$$l \cdot (f+h) - l \cdot f \in F^{s+1}_{\alpha,\beta}M$$

from lemma 2.24(iii). Taking $s = k+1$ the hypotheses of proposition 1.3 hold (we put W equal to the projection of $F^{k+1}_{\alpha,\beta}M$ into $J^{k+1}(n,p)$) and the result follows. □

Remark 2.26. This is a finite-dimensional problem in the weighted $(k+1)$-jet space $F^0_{\alpha,\beta}M/F^{k+2}_{\alpha,\beta}M$. The theorem can be rephrased as 'any germ g with the same (weighted) k-jet as f is $F^1\mathcal{G}$-equivalent to a germ with $(k+1)$-jet of the form $j^{k+1}f + j^{k+1}t$, for some $t \in T$'.

The following weighted determinacy result from [9, theorem 5.2] follows as a corollary.

Corollary 2.27. *Let $\mathcal{G}$, α and β be as in theorem 2.25. Then a map-germ $f \in F^0_{\alpha,\beta}M$ is (weighted) k-$\mathcal{G}$-determined (i.e. $F^{k+1}_{\alpha,\beta}M$-$\mathcal{G}$-determined) if*

$$F^{k+1}_{\alpha,\beta}M \subset L(F^1\mathcal{G}) \cdot f.$$

We now suppose that our germ f is weighted homogeneous, and obtain a different type of result.

Theorem 2.28. *Let $\mathcal{G}$ be one of the standard Mather groups $\mathcal{R}$, $\mathcal{L}$, $\mathcal{A}$, $\mathcal{C}$ or $\mathcal{K}$ and $\alpha = (\alpha_1, \ldots, \alpha_n)$, $\beta = (\beta_1, \ldots, \beta_p)$ be sequences of weights as above. Suppose that f is weighted homogeneous of degree r, and take $s > r$. Let T be a subspace of $F^{r+1}_{\alpha,\beta}M$ such that*

$$F^{r+1}_{\alpha,\beta}M \subset T + L(F^1\mathcal{G}) \cdot f + F^{s+1}_{\alpha,\beta}M.$$

Then any germ g with $f-g \in F^{r+1}_{\alpha,\beta}M$ is $F^1\mathcal{G}$-equivalent to a map-germ of the form $f+t+\phi$ where $t \in T$ and $\phi \in F^{s+1}_{\alpha,\beta}M$.

Proof. The proof is by induction on s. The case $s = r$ is trivial (and remark that the case $s = r+1$ is just theorem 2.24). Suppose the result is established for $s = m$, and the hypotheses hold for $s = m+1$. We can suppose that T is generated as a $\boldsymbol{K}$-vector space by polynomial maps of degree D with $r+1 \leqslant D \leqslant m+1$. We write the generators of degree $\leqslant m$ as $\{e_i\}$ and those of degree $m+1$ as $\{d_j\}$. Note that the hypotheses hold for $s = m$ and $T' = \boldsymbol{K}\{e_i\}$ so that given g as above we find it is $F^1\mathcal{G}$-equivalent to something of the form $f + \sum \lambda_i e_i + \psi$ where $\lambda_i \in \boldsymbol{K}$ and $\psi \in F^{m+1}M$. We claim that

$$F^{m+1}M \subset L(F^1\mathcal{G}) \cdot (f + \sum \lambda_i e_i) + \boldsymbol{K}\{d_j\} + F^{m+2}M.$$

For if $\gamma \in F^{m+1}M$ we can write $\gamma = \xi \cdot f + \sum \mu_i d_i + \delta$, where ξ is a weighted homogeneous vector field of degree $m+1-r$, $\mu_i \in \boldsymbol{K}$ and $\delta \in F^{m+2}M$. Now $\xi \cdot (f + \sum \lambda_i e_i) - \xi \cdot f \in F^{m+2}M$ by lemma 2.24(iii) as $\sum \lambda_i e_i \in F^{r+1}M$, and the claim holds. We can now apply theorem 2.25 to $f + \sum \lambda_i e_i$ and $f + \sum \lambda_i e_i + \psi$ and deduce that g is $F^1\mathcal{G}$-equivalent to a germ of the form $f + \sum \lambda_i e_i + \sum \mu_i d_i + \phi$ where $\phi \in F^{m+2}M$. □

Remark 2.29. The theorem can be rephrased as 'any germ g with the same (weighted) r-jet as f is $F^1\mathcal{G}$-equivalent to a germ with s-jet of the form $j^s f + j^s t$, for some $t \in T$'.

Example 2.30. In the case of functions ($p = 1$) this result is due to Arnold [3], where many applications are given. We now consider some examples when $p \geqslant 2$.

(i) Consider the classification of $\mathcal{K}$-unimodular germs $\boldsymbol{C}^2, 0 \to \boldsymbol{C}^2, 0$ made by Dimca and Gibson in [12]. One of the germs they consider is $(x^3 + y^5, x^2y)$, which is of finite $\mathcal{K}$-codimension. If we assign weights $(\alpha_1, \alpha_2) = (5, 3)$, $(\beta_1, \beta_2) = (15, 13)$ then this germ is weighted homogeneous. Moreover, the vector $(xy^4, 0)$ spans the relevant quotient space, so from theorem 2.28 we deduce that any germ with this weighted 0-jet is equivalent to something of the form $(x^3 + y^5 + axy^4, x^2y)$, and indeed an obvious scaling can be made to reduce a to 0 or 1. (In [12] a further possible germ with the given 0-jet is given, namely $(x^3 + y^5, x^2y + y^5)$, but one can show that this is $\mathcal{K}$-equivalent to $(x^3 + y^5 + xy^4, x^2y)$.) Similarly the germ $(x^3 + 12xy^3, x^2y + y^4)$ is of finite $\mathcal{K}$-codimension, and is weighted homogeneous if we assign weights $(\alpha_1, \alpha_2) = (3, 2)$, $(\beta_1, \beta_2) = (9, 8)$. The corresponding quotient space this time is spanned by $(y^5, 0)$, so any germ with this weighted 0-jet is $\mathcal{K}$-equivalent to something of the form $(x^3 + 12xy^3 + ay^5, x^2y + y^4)$. By scaling we can reduce a to 0 or 1 and obtain two normal forms. (In [12] a third normal form is given, namely $(x^3 + 12xy^3 + y^6, x^2y + y^4)$, but this is $\mathcal{K}$-equivalent to $(x^3 + 12xy^3, x^2y + y^4)$.) These facts are also given in [26, p 290].

(ii) Consider the $\mathcal{A}$-classification of map-germs $\boldsymbol{C}^2, 0 \to \boldsymbol{C}^2, 0$ and define $f(x, y) = (x, x^k y + y^4)$. If we assign weights $(\alpha_1, \alpha_2) = (3, k)$, $(\beta_1, \beta_2) = (3, 4k)$ this germ is weighted homogeneous. A straightforward calculation using theorem 2.28 shows that any germ with weighted 0-jet equal to f can be written in the form

$$\left(x, x^k y + y^4 + \sum_{2k \geqslant j \geqslant 2k/3} a_j x^j y^2\right).$$

It follows from the work of Damon or Gaffney that these families are right–left topologically trivial (see [14]).

3. Implementation on a computer

Let $\mathcal{G}$ be a subgroup of $\mathcal{K}$, typically one of the standard Mather groups $\mathcal{R}$, $\mathcal{L}$, $\mathcal{A}$, $\mathcal{C}$ or $\mathcal{K}$ or more generally one of Damon's geometric subgroups of $\mathcal{K}$. We seek to list orbits of finitely determined germs $f \in \mathcal{M}_n\mathcal{E}(n, p)$ under the action of $\mathcal{G}$, choosing suitable normal forms as representatives. Classification is done inductively at the jet-level via the method of complete transversals. As we have seen, establishing the degree of determinacy of a germ is essentially a special case of this problem (we have to show that successive complete transversals are empty). Both calculations therefore become ones in linear algebra; the main objective being to calculate the tangent space to the orbit of the germ in the jet space.

The procedure the computer program utilizes is summarized below. We use the above ideas to obtain an effective classification procedure where we classify jets inductively until determined jets result (or pre-selected upper bounds on moduli or codimension are reached).

Step 1. For a given germ f, jet space of degree k and group $\mathcal{G}$, calculate the tangent space to the orbit of f in the jet space of degree k, that is $L(J^k\mathcal{G})\cdot(j^k f)$ in $J^k(n, p)$. Specifically, a spanning set for this tangent space as a vector subspace of $J^k(n, p)$ is calculated. The algorithm is essentially the same as that followed if one were doing the calculation by hand. For example, in a standard $\mathcal{R}$ classification, using the $\mathcal{R}_1$ classification technique say, we first obtain the vectors which generate the $\mathcal{R}$ tangent space as an $\mathcal{E}_n$-module. These are multiplied by all monomials of degree 2 and higher in the source variables until we obtain jets whose initial degree is greater than the given degree k.

Algorithms to determine the tangent spaces to all of the standard Mather groups are available in the program. In addition, the vector fields which give the generators to the $\mathcal{R}$ tangent space are also user-defined allowing one to restrict to subgroups of $\mathcal{R}$ which, for example, preserve a given variety.

In the case of $\mathcal{A}$-classification, one must work with the $\mathcal{A}_1$ tangent space with appropriate 'nilpotent' terms included. The $\mathcal{A}_1$ tangent space at f is given (in the notation of Mather) by

$$L\mathcal{A}_1 \cdot f = tf(\mathcal{M}_n^{\text{source_power}}\theta_n) + wf(\mathcal{M}_p^{\text{target_power}}\theta_p)$$

where `source_power` $= 2$, `target_power` $= 2$. However, by allowing the user to specify the variables `source_power` and `target_power` one can work with the extended 'group' $\mathcal{A}_e$ (for example, to calculate versal unfoldings, [19]) or the $\mathcal{A}$ group itself (for example, to check the hypotheses of the Mather lemma). Similar observations apply to the other groups. The classification package is therefore capable of performing much more than complete transversal and determinacy calculations.

Step 2. The spanning set calculated in step 1 is reduced to echelon form using Gaussian elimination. This is done by ordering the monomial vectors in $J^k(n, p)$, extracting the coefficients from the vectors obtained in step 1, and then reducing the matrix whose rows consist of these coordinate vectors using Gaussian elimination. (We refer to the remarks below, however.) This gives a canonical basis for the tangent space.

Step 3. A basis C for the complementary (normal) space to the tangent space is calculated. That is, the independent set obtained in step 2 is extended to one of full rank in $J^k(n, p)$. In the $\mathcal{A}_e$ and $\mathcal{A}$ cases (for example) converting the elements of C back to monomial vectors gives the terms required in a versal unfolding and the corresponding codimension

(for determined jets). In the 'unipotent' and $\mathcal{A}_1$ cases (for example) this gives a basis for a complete transversal. By ordering the monomial jets in $J^k(n, p)$ so that those of degree k correspond to the later columns of the matrix, those monomial vectors in C of degree k will form a basis for a complete transversal. Of course, in the 'unipotent' case not only must we arrange for the latter columns to correspond to the degree k monomials but we must order such monomials amongst themselves using the order induced by the corresponding nilpotent filtration.

Step 4. Steps 1–3 form the major part of the calculation and are performed by one procedure. The bases for the tangent and normal spaces are stored for global access by other routines. Step 4 deals with output. A number of procedures are associated with step 4 and perform such functions as displaying the bases and testing whether a given set of vectors is independent from the tangent space (such questions arise in checking the hypotheses of the Mather lemma and in moduli detection).

Remark 3.1. The above steps describe the basic algorithms and functionality of our package. Most of the technical points have, of course, been avoided and we refer to [17] and [18] for the full details. However, there are a few points we would like to mention before moving on.

(i) Writing the package in a symbolic language such as `Maple` allows great flexibility. One notable advantage is that parameters (such as moduli) may be present in the jets we work with. The matrix created in step 2 will then contain polynomial entries and we must then take care during the Gaussian elimination routine. We choose numeric pivotal elements (in this context meaning 'constant polynomials') where possible, but when we are forced to choose a non-constant polynomial pivotal element *no* division is performed on the chosen row to reduce the pivot to unity. (Division can still be performed when using the pivot to reduce the rest of the column to zero, in contrast with standard 'fraction-free' Gaussian elimination where the matrix entries rapidly blow-up to large expressions). We then make a list of all the non-numeric pivotal elements; these will, in general, be rational functions in the moduli parameters. For most members of the family f, each corresponding to specific values of the moduli, the rational functions do not vanish and the elimination is valid. However, we must inspect each rational function in turn obtaining conditions on the moduli for which the elimination breaks down. To investigate the exceptional behaviour the solutions to these conditions must be substituted back into the family. The list of rational functions is stored for global access and may be examined by a procedure associated with step 4.

(ii) A major problem with this approach is the size of the matrices produced in step 2. However, these matrices are highly sparse and, as numerous examples prove, can be reduced relatively quickly. The success of the method in quite complicated classifications shows that this is not a major problem. (It appears to be essentially unavoidable anyway!)

Many techniques are used in the elimination process to exploit the sparsity of such matrices and improve the efficiency of the package. We will mention two important features; first 'indexed Gaussian elimination'. Creating the matrix of coefficients referred to in step 2 is wasteful on memory and CPU time. The data is already stored in a compact form as a table of tangent vectors; each entry is just a vector of polynomials (truncated to degree k) and therefore stores the sparse data (the non-zero coefficients) in a minimal way. We still use the concept of a matrix of coefficients so that Gaussian elimination can be used to produce a basis. The idea is to use reference tables which, for a given row and column (i, j), index the appropriate entry in the corresponding matrix, as a coefficient of the ith

polynomial vector in our spanning set. The row reduction operations performed in Gaussian elimination are now achieved by polynomial addition—a very efficient process in `Maple` which uses the internal functions (and replaces a large number of 'get coefficient' and '$0 -$ (scale factor) $* \, 0 = 0$' operations which inevitably occur during row reduction of a sparse matrix). This method, on average, proved to be two to three times faster, using three to four times less memory than methods which extract an explicit matrix of coefficients.

Another important feature is the exploitation of the symmetry present in the left tangent space. This applies to both the $\mathcal{L}$ and $\mathcal{C}$ cases; we give details in the $\mathcal{L}$ case. In standard coordinates the left tangent space is given by

$$f^*(m_p^{\text{target_power}}) \cdot \{e_1, \ldots, e_p\}.$$

We create a spanning set for the ideal $f^*(m_p^{\text{target_power}})$ as a vector subspace of $J^k(n, 1)$ and reduce this to echelon form using (indexed) Gaussian elimination. We then produce a spanning set for the full left tangent space by stacking together p copies of the resulting 'matrix'. The important point is that we can do this in such a manner as to create a spanning set for the full left tangent space which is *already* in echelon form and therefore requires no further elimination. The reduction in computational overhead is clear. Finally, in an $\mathcal{A}$-classification, this basis for the left tangent space is adjoined with a spanning set for the right tangent space and the resulting 'matrix' reduced to echelon form. Again further features can be exploited and a full-blown Gaussian elimination routine is not required. The final elimination algorithm, though somewhat more complicated than the standard Gaussian elimination algorithm, is more efficient. We refer to [17] and [18] for full details and elaboration on all of the above points.

(iii) Variants on the package to deal with weighted homogeneous filtrations and multigerm classifications have also been developed [17, 18, 15].

4. An extended example: classification of function germs on discriminant varieties

In this section we show how our methods yield a classification of function-germs on discriminant varieties generalizing those given by Arnold in [1], and [3]. This is our major example of a classification (most of the substantial applications of this paper will appear elsewhere). We suppose $\mathcal{D}, 0 \subset \boldsymbol{C}^p, 0$ is a discriminant variety and study the classification of functions $\boldsymbol{C}^p, 0 \to \boldsymbol{C}, 0$ under $\mathcal{R}(\mathcal{D})$-equivalence, the restriction of $\mathcal{R}$-equivalence to source coordinate changes which preserve the variety. We consider the cases where $\mathcal{D}$ is the discriminant of the A_2 and A_3 singularities, extending the results of Arnold. Several other simple singularities are discussed briefly; in particular, the cases where $\mathcal{D}$ is the discriminant of the A_4, D_4, D_5, D_6, and the E_k singularities.

We will set the scene and state the results; details of the actual calculations performed are summarized in [17]. We shall work in the complex analytic case. In this case the module of germs of vector fields tangent to $\mathcal{D}$, $\Theta(\mathcal{D})$, is finitely generated as a module over $\mathcal{O}_p$; indeed it is a free module and there is an algorithm due to Saito, [30], for calculating the generators. Our results also provide a classification in the real smooth case (since integration of the Saito vector fields yields diffeomorphisms preserving the real discriminant). Classifications in the real case are important in geometrical applications; see [1, 5].

This section extends the scope of our techniques and demonstrates them in a somewhat more complicated setting. We need to work with the module of vector fields tangent to the given discriminant, $\Theta(\mathcal{D})$. An algorithm which implements Saito's method of construction was developed in [17]. Even for moderately low codimension singularities the Saito vector fields are rather complicated, requiring several pages to write down. Consequently the

use of a computer algorithm to generate them is a prerequisite for any classification of functions on such discriminants. The computer package described in section 3 is fully exploited throughout. There is a natural choice of weights to assign to the coordinates on $\boldsymbol{C}^p$ and these cases provide good applications for the weighted homogeneous version of the complete transversal theorem.

4.1. Complete transversals and determinacy results

For a general discussion of determinacy of functions on analytic varieties we refer to [10]. Let $X, 0 \subset \boldsymbol{C}^p, 0$ be the germ of a reduced analytic subvariety of $\boldsymbol{C}^p$ at 0. We shall consider analytic function germs $\boldsymbol{C}^p, 0 \to \boldsymbol{C}, 0$ and say two germs are equivalent if one can be obtained from the other by source coordinate changes which preserve X. The group $\mathcal{R}(X)$ is one of Damon's 'geometric subgroups' of $\mathcal{A}$ (see [11]) and the following determinacy result holds.

Theorem 4.1. *A germ $h : \boldsymbol{C}^p, 0 \to \boldsymbol{C}, 0$ is finitely $\mathcal{R}(X)$-determined if the ideal*

$$J_X(h) = \{\delta \cdot h : \delta \in \Theta(X)\}$$

in $\mathcal{O}_p$ contains some power of the maximal ideal $\mathcal{M}_p$.

If in addition the vector fields $\delta \in \Theta(X)$ vanish at 0 then $\mathcal{M}_n^{k+1} \subset \mathcal{M}_n \cdot J_X(h)$ implies that h is k-$\mathcal{R}(X)$-determined.

For discriminants of $\mathcal{K}$-miniversal unfoldings the tangent vector fields *do* all vanish at the origin. In many situations the vector fields have a natural weighting and we can improve on this criterion by introducing a weighted filtration and using 'more' of the module $\Theta(X)$ in the calculation. The following can be deduced from [4] for example.

Proposition 4.2. *Let $f : \boldsymbol{C}^n, 0 \to \boldsymbol{C}, 0$ be a finitely determined analytic germ and $\mathcal{D}$ its discriminant. Suppose further that f is weighted homogeneous of weight d with respect to the system of weights $(\bar{w}_1, \ldots, \bar{w}_n)$. Let $\{g_1, \ldots, g_p\}$ be a basis of monomials for $\mathcal{O}_n/J(f)$ and ordered such that*

$$\mathrm{wt}(g_1) \geqslant \mathrm{wt}(g_2) \geqslant \cdots \geqslant \mathrm{wt}(g_{p-1}) > \mathrm{wt}(g_p) = 0$$

(so choose $g_p = 1$). Define α_i to be $\mathrm{wt}(g_i)$. Let $(u_1, \ldots, u_p)$ be coordinates on $\boldsymbol{C}^p$ and assign weights $w = (w_1, \ldots, w_p)$ where $w_i = d - \alpha_i$. Let a_{ij} be defined by

$$F.g_j = \sum_i a_{ij} g_i \bmod J(F)$$

and θ_j be the Saito vector fields which generate $\Theta(\mathcal{D})$:

$$\theta_j = \sum_i a_{ij} \partial/\partial u_i$$

as described in [4]. Then

(i) the a_{ij} are weighted homogeneous and

$$\mathrm{wt}(a_{ij}) = w_i + \alpha_j = d - \alpha_i + \alpha_j$$

(ii) the vector fields θ_j are weighted homogeneous and

$$\mathrm{wt}(\theta_j) = \alpha_j.$$

(That is, $\theta_j \cdot g \in F_w^{t+\alpha_j}\mathcal{O}_p$ for all $g \in F_w^t\mathcal{O}_p$ and all t.)

Remark 4.3. We will consider the cases where $\alpha_i < d$ for all i (this implies f must be one of the simple singularities) so that the weights w_i are positive; see [4]. The vector field θ_p of weight 0 is called the *Euler vector field*.

It is now natural to formulate complete transversal and determinacy results using a weighted filtration. We will require the following technical lemma.

Lemma 4.4. *(i) Let $X, 0 \subset \boldsymbol{C}^p, 0$ be the germ of an analytic variety, then $L(\mathcal{R}(X)) = \Theta(X)$ (using the definition given in [9, p 540]).*

(ii) Let $\mathcal{D}, 0 \subset \boldsymbol{C}^p, 0$ be the discriminant of a function-germ f. Suppose further that f is right equivalent to a weighted homogeneous function. Then for all s we have $J^s(L\mathcal{R}(\mathcal{D})) \subset L(J^s\mathcal{R}(\mathcal{D}))$.

Proof. (i) Let $\Phi : \boldsymbol{C}^p \times \boldsymbol{C}, 0 \to \boldsymbol{C}^p, 0$ be an analytic germ with $\Phi(x, 0) = x$ for all $x \in \boldsymbol{C}^p$ and $\Phi(X, t) = X$ for small t. Consider the vector field $x \mapsto \partial\Phi(x, t)/\partial t|_{t=0}$, also denoted $\partial\Phi_t/\partial t|_{t=0}$. Clearly this is tangent to X, i.e. lies in $\Theta(X)$. Conversely, any vector field in $\Theta(X)$ can be integrated to give a flow Φ as above.

(ii) If $\xi \in J^s(L\mathcal{R}(\mathcal{D}))$ then we can find some $\tilde{\xi} \in L\mathcal{R}(\mathcal{D})$ with $j^s\tilde{\xi} = \xi$. In other words for some flow $\Phi : \boldsymbol{C}^p \times \boldsymbol{C}, 0 \to \boldsymbol{C}^p, 0$ as in (i) we have $\tilde{\xi} = \partial\Phi_t/\partial t|_{t=0}$. We know that all vector fields in $\Theta(\mathcal{D})$ vanish at the origin, so for all small t, $\Phi_t : \boldsymbol{C}^p, 0 \to \boldsymbol{C}^p, 0$ and $\Phi_t(X) = X$. That is $t \mapsto j^s(\Phi_t)$ is a path in $J^s\mathcal{R}(\mathcal{D})$ so $\partial(j^s\Phi_t)/\partial t|_{t=0}$ lies in $L(J^s\mathcal{R}(\mathcal{D}))$. But $j^s(\partial\Phi_t/\partial t|_{t=0}) = \partial(j^s\Phi_t)/\partial t|_{t=0}$ so the result holds. □

Theorem 4.5. *Assume the hypotheses and notation of proposition 4.2. Let $\{F^r_w\mathcal{O}_p\}$ be the weighted filtration of $\mathcal{M}_p$, which we will abbreviate to $\{F^r\mathcal{O}_p\}$. The weighted filtration $\{F^r\mathcal{R}\}$ induces a filtration of $\mathcal{R}(\mathcal{D})$ and the results of section 2.3 hold. Define*

$$L = \langle\theta_1, \ldots, \theta_{p-1}\rangle + F^1\mathcal{O}_p.\langle\theta_p\rangle \subset L\mathcal{R}(\mathcal{D}).$$

Let $h \in \mathcal{M}_p$, $k \geqslant 1$ and T be a subspace of $F^{k+1}\mathcal{O}_p$ such that

$$F^{k+1}\mathcal{O}_p \subset T + L \cdot h + F^{k+2}\mathcal{O}_p$$

then any germ h_1 with $h_1 - h \in F^{k+1}\mathcal{O}_p$ is $F^1\mathcal{R}(\mathcal{D})$ equivalent to $h + t + \phi$ for some $t \in T$ and $\phi \in F^{k+2}\mathcal{O}_p$.

Proof. The proof follows from proposition 1.3 in a similar way to that of theorem 2.25. We will only sketch the proof, highlighting the main points. The main question now is, do the resulting coordinate changes in $\boldsymbol{C}^p$ preserve $\mathcal{D}$? From lemma 4.4 we have $J^s(L\mathcal{R}(\mathcal{D})) \subset L(J^s\mathcal{R}(\mathcal{D}))$ for all s; this is the important result. (The problem was avoided in theorem 2.25 by appealing to the fact that $\mathcal{G}$ is jet closed. In the present setting it is not clear if $\mathcal{R}(\mathcal{D})$ is jet closed with respect to the weighted filtration.) It follows that $J^{k+1}L \subset J^{k+1}(L\mathcal{R}(\mathcal{D}))$ is a Lie subalgebra of $L(J^{k+1}\mathcal{R}(\mathcal{D}))$ so there is a Lie subgroup G of $J^{k+1}\mathcal{R}(\mathcal{D})$ with Lie algebra $J^{k+1}L$. The result therefore follows from proposition 1.3 once we have verified that for $f \in F^1\mathcal{O}_p$, $g \in F^s\mathcal{O}_p$ and $l \in L$

$$l \cdot (f + g) - l \cdot f \in F^{s+1}\mathcal{O}_p.$$

But this follows from the fact that the vector fields θ_j are weighted homogeneous of positive weight for $j = 1, \ldots, p-1$ (by lemma 4.2) and that they act linearly. □

The following determinacy result is a corollary to theorem 4.5. It is a major improvement over the determinacy result in theorem 4.1 where one needs to use the space $F^1\mathcal{O}_p.\langle\theta_1, \ldots, \theta_p\rangle$ in place of L.

Corollary 4.6. *With the notation of theorem 4.5, a germ $h : \boldsymbol{C}^p, 0 \to \boldsymbol{C}, 0$ is k-$\mathcal{R}(\mathcal{D})$-determined if*

$$F^{k+1}\mathcal{O}_p \subset L \cdot h.$$

As usual to carry out the determinacy calculations effectively we must reduce this to a finite dimensional problem.

Corollary 4.7. *Define w_{max} to be the maximum of the weights $w_1, \ldots, w_p$. A germ $h : \boldsymbol{C}^p, 0 \to \boldsymbol{C}, 0$ is k-$\mathcal{R}(\mathcal{D})$-determined if*

$$F^{k+1}\mathcal{O}_p \subset L \cdot h + F^{k+1+w_{max}}\mathcal{O}_p.$$

Proof. Since $\mathcal{O}_p$ is a local ring we can apply the Nakayama lemma to see that the condition in corollary 4.6 is equivalent to

$$F^{k+1}\mathcal{O}_p \subset L \cdot h + \mathcal{M}_p.F^{k+1}\mathcal{O}_p.$$

Now, by definition, the ideal $F^{k+1+w_{max}}\mathcal{O}_p$ is generated by the monomials in $\mathcal{O}_p$ of weight $\geqslant k + 1 + w_{max}$. Such monomials can therefore be written in the form $u_i m$ for $1 \leqslant i \leqslant p$ and m a monomial of weight $\geqslant k + 1$. Thus, $u_i m \in \mathcal{M}_p.F^{k+1}\mathcal{O}_p$ so $F^{k+1+w_{max}}\mathcal{O}_p \subset \mathcal{M}_p.F^{k+1}\mathcal{O}_p$ and the result follows. □

To check determinacy we can therefore work in the $(k + w_{max})$-jet space and perform the calculation using a computer. Indeed, suppose we consider h as a k-jet and calculate the complete transversals of weight $k + 1$ to weight $k + w_{max}$ using the computer. If these are all found to be empty then the condition in corollary 4.7 holds and h is k-$\mathcal{R}(\mathcal{D})$-determined. (We can, of course, improve matters by using $\mathcal{M}_p.F^{k+1}\mathcal{O}_p$ instead of $F^{k+1+w_{max}}\mathcal{O}_p$, but this was found to be unnecessary in practice, and hard to implement on a computer.)

Of course, one class of coordinate change denied us by the complete transversal method is scaling. Generally such coordinate changes will not preserve the discriminant $\mathcal{D}$. There is, however, one scale change that clearly does.

Proposition 4.8. *Assume the hypotheses and conventions of proposition 4.2. Then for $t \in \boldsymbol{C}$, $t \neq 0$, the map-germ $\boldsymbol{C}^p, 0 \to \boldsymbol{C}^p, 0$ defined by*

$$(u_1, \ldots, u_p) \mapsto (t^{w_1}u_1, \ldots, t^{w_p}u_p)$$

is an element of the group $\mathcal{R}(\mathcal{D})$.

4.2. Classification of function-germs on the A_2 (cusp) and A_3 (swallowtail) discriminants

In this section we classify all function-germs on the A_2 (cusp) and A_3 (swallowtail) discriminant varieties $\mathcal{D}$ under $\mathcal{R}(\mathcal{D})$-equivalence and up to modality 2. This extends the results of [1] where the same weighted filtration was employed. Some families of higher modality occur naturally in the classification and we include these too. We will use $a, b, c, \ldots$ to denote moduli.

A weighted homogeneous normal form for the A_k singularity is $f(x) = x^{k+1}$. Then $J(f) = \langle x^k \rangle$ and we take $g_1 = x^{k-1}$, $g_2 = x^{k-2}, \ldots, g_k = 1$ as a basis for $\mathcal{O}_n/J(f)$, and assign weights $(2, 3, \ldots, k+1)$ to the coordinates $(u_1, \ldots, u_k)$ on $\boldsymbol{C}^k$, as in proposition 4.2. The techniques described in section 4.1 apply and we obtain the following classifications.

Theorem 4.9. *Every function-germ $h : \boldsymbol{C}^2, 0 \to \boldsymbol{C}, 0$ on the cusp discriminant variety $\mathcal{D}$ of $\mathcal{R}(\mathcal{D})$-modality $\leqslant 2$ is $\mathcal{R}(\mathcal{D})$-equivalent to one of the following finitely determined germs. The first germs of modality $\geqslant 3$ to occur during the classification are given in the second section of the table; they occur as the initial singularities in the stated series. The codimension refers to the dimension of $\mathcal{O}_k/J_X(h)$ as a $\boldsymbol{C}$-vector space; u_1 is the only stable singularity.*

Singularity	*Determinacy degree*	*Codim*
u_1	*2*	*1*
u_2	*3*	*2*
$u_1^2 + au_2^n$	$3n, \quad n \geqslant 2, \quad a \neq 0$	$n+2$
$u_1u_2 + au_1^3$	*6*	*5*
$u_1^3 + au_2^2 + bu_1u_2^2$	$8, \quad 4a \neq 27, \quad a \neq 0$	*7*
$u_1^3 + \frac{27}{4}u_2^2 + au_1^nu_2 + bu_1^{n+1}u_2$	$2n+5, \quad n \geqslant 2, \quad a \neq 0$	$2n+4$
$u_1^3 + \frac{27}{4}u_2^2 + au_1^n + bu_1^{n+1}$	$2n+2, \quad n \geqslant 4, \quad a \neq 0$	$2n+1$
$u_2^2 + au_1^n + bu_1^{n+1}$	$2n+2, \quad n \geqslant 4, \quad a \neq 0$	$n+4$
$u_1^3 + au_1u_2^n + bu_2^{n+1} + cu_2^{n+2}$	$3n+6, \quad n \geqslant 2, \quad a, b \neq 0$[a]	$2n+5$
$u_1^3 + au_2^n + bu_1u_2^n$	$3n+2, \quad n \geqslant 3, \quad a \neq 0$	$2n+3$
$u_1^2u_2 + au_1^4 + bu_2^n + cu_2^{n+1}$	$3n+3, \quad n \geqslant 3, \quad b \neq 0$	$n+7$

[a] *For the case $n = 2$ the condition $b \neq 0$ needs to be replaced by $4a^3 + 27b^2 \neq 0$.*

Theorem 4.10. *Every function-germ $h : \boldsymbol{C}^3, 0 \to \boldsymbol{C}, 0$ on the swallowtail discriminant variety $\mathcal{D}$ of $\mathcal{R}(\mathcal{D})$-modality $\leqslant 2$ is $\mathcal{R}(\mathcal{D})$-equivalent to one of the following finitely determined germs. The first germ of modality $\geqslant 3$ to occur during the classification is given in the second section of the table. The codimension refers to the dimension of $\mathcal{O}_k/J_X(h)$ as a $\boldsymbol{C}$-vector space; u_1 is the only stable singularity.*

Singularity	*Determinacy degree*	*Codim*
u_1	*2*	*1*
$u_2 + au_1^n$	$2n, \quad n \geqslant 2, \quad a \neq 0$	$n+1$
$u_3 + au_1^n + bu_1^{n+1}$	$2n+2, \quad n \geqslant 2, \quad a \neq 0$[a]	$n+3$
$u_3 - \frac{1}{4}u_1^2 + au_1^n + bu_1^{n+1}$	$2n+2, \quad n \geqslant 3, \quad a \neq 0$	$n+3$
$u_3 + \frac{1}{12}u_1^2 + au_1^nu_2 + bu_1^{n+1}u_2$	$2n+5, \quad n \geqslant 1, \quad a \neq 0$	$2n+4$
$u_3 + \frac{1}{12}u_1^2 + au_1^n + bu_1^{n+1}$	$2n+2, \quad n \geqslant 3, \quad a \neq 0$	$2n+1$
$u_1^2 + bu_2^2 + cu_2u_3 + du_3^2 + eu_3^3$	*12,* $\quad b, c, 4bd - c^2 \neq 0$	*8*

[a] *For the case $n = 2$ the extra conditions $12a - 1 \neq 0$ and $4a + 1 \neq 0$ are required.*

4.3. Classification of function-germs on the A_k, D_k and E_k discriminants

Out of interest we investigate the classification of function-germs on discriminants of the other simple singularities. We consider the cases A_4, D_4, D_5, D_6, E_6, E_7 and E_8. Generally the first singularity to occur after the stable singularity is of high modality. Our results demonstrate that even during the initial stages of such classifications complicated singularities arise.

All the calculations were performed by computer, including the far from trivial calculation of the vectors fields tangent to the given discriminant. We use the weighted homogeneous normal forms f, and corresponding monomial basis for $\mathcal{O}_n/J(f)$, indicated in table 3. Coordinates on $\boldsymbol{C}^p$ are denoted $(u_1, \ldots, u_p)$ and we assign weights according to proposition 4.2; see table 4. The initial stages of each classification are shown in table 5. The codimension refers to the dimension of $\mathcal{O}_p/J_X(h)$ as a $\boldsymbol{C}$-vector space, and 'det' to the determinacy degree. In each case u_1 is the only stable singularity. The a_i denote moduli and the determinacy degree for the non-simple cases holds for generic values of the moduli (more precisely, on the complement of a proper algebraic subvariety of the corresponding affine space).

Remark 4.11. The above results lead us to make the following conjectures.

Table 3. Normal forms and generators for the normal space.

Type	Normal form f	Basis for $\mathcal{O}_n/J(f)$
D_4	$x^2y + y^3$	$\{y^2, x, y, 1\}$
D_5	$x^2y + y^4$	$\{y^3, y^2, x, y, 1\}$
D_6	$x^2y + y^5$	$\{y^4, y^3, x, y^2, y, 1\}$
E_6	$x^3 + y^4$	$\{xy^2, xy, y^2, x, y, 1\}$
E_7	$x^3 + xy^3$	$\{y^4, y^3, xy, y^2, x, y, 1\}$
E_8	$x^3 + y^5$	$\{xy^3, xy^2, y^3, xy, y^2, x, y, 1\}$

Table 4. Weights for coordinates on $\boldsymbol{C}^n$ and $\boldsymbol{C}^p$.

	Weights										
Type	x	y	u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	Total weight
A_4	1	—	2	3	4	5	—	—	—	—	5
D_4	1	1	1	2	2	3	—	—	—	—	3
D_5	3	2	2	4	5	6	8	—	—	—	8
D_6	4	2	2	4	6	6	8	10	—	—	10
E_6	4	3	2	5	6	8	9	12	—	—	12
E_7	3	2	1	3	4	5	6	7	9	—	9
E_8	5	3	1	4	6	7	9	10	12	15	15

Table 5. Classification of singularities on $\mathcal{D}, 0 \subset \boldsymbol{C}^p, 0$.

Type	Singularity	Det	Codim
A_4	u_1	2	1
	$u_2 + a_1u_1^2$	4	3
	$u_3 + a_1u_1^2 + a_2u_1u_2 + a_3u_1^3 + a_4u_1^4$	8	7
D_4	u_1	1	1
	$u_2 + a_1u_3 + a_2u_1^2 + a_3u_1^3 + a_4u_1^4$	4	6
D_5	u_1	2	1
	$u_2 + a_1u_1^2 + a_2u_3 + a_3u_1^3 + a_4u_1^4 + a_5u_1^5$	10	7
D_6	u_1	2	1
	$u_2 + a_1u_1^2 + a_2u_3 + a_3u_1^3 + a_4u_1^4 + a_5u_1^5 + a_6u_1^6$	12	8

(i) Let $\mathcal{D}$ be a discriminant variety of any simple singularity with the exception of A_2. Then the classification of function-germs $h : \boldsymbol{C}^p, 0 \to \boldsymbol{C}, 0$ under $\mathcal{R}(\mathcal{D})$-equivalence yields only one simple germ, the stable germ u_1. (In the A_2 case we obtain two simples, namely u_1 and u_2.)

(ii) For the D_k discriminant the next least degenerate singularity after the stable singularity u_1, that is, the singularity with non-zero u_2 coefficient, can be put in the form $u_2 + a_1u_3 + a_2u_1^2 + a_3u_1^3 + \cdots + a_ku_1^k$ which is $2k$-determined and of codimension $k + 2$.

Remark 4.12. In the E_6, E_7 and E_8 cases, the next least degenerate singularity after the stable singularity u_1 (in each case, the singularity with non-zero u_1^2 coefficient) has modality greater than 6, 7 and 8, respectively.

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